SCF Assignment 4 — BE22B008

Exercise 1(c)

$$\int_{X,Y} f(x,y)\, dx\, dy = \frac{1}{2\pi} e^{-(x^2+y^2)/2}\, dx\, dy$$

Substituting, $x = \sqrt{2\rho}\cos\theta$

$y = \sqrt{2\rho}\sin\theta$

$\Rightarrow x^2 + y^2 = 2\rho$

$$\int_{X,Y} dx\, dy = \frac{1}{2\pi} e^{-\rho}\, d\rho\, d\theta$$

$$= \left(e^{-\rho} d\rho\right)\left(\frac{1}{2\pi} d\theta\right)$$

$$= \int_R (\rho)\, d\rho \cdot \int_\Theta (\theta)\, d\theta$$

Exercise II

Means

$X' = aX + bY$ and $Y' = cX + dY$

$\mu_1' = \mathbb{E}[X'] = a\mu_1 + b\mu_2$

$\mu_2' = \mathbb{E}[Y'] = c\mu_1 + d\mu_2$

Covariance Matrix

Since X and Y are independent with equal variance σ^2, we have

$Cov(X, Y) = 0$; $Var(X) = Var(Y) = \sigma^2$

The centred variables are $X - \mu_1$ and $Y - \mu_2$, so...

$X' - \mu_1' = a(X - \mu_1) + b(Y - \mu_2)$

$Y' - \mu_2' = c(X - \mu_1) + d(Y - \mu_2)$

• $Var(X') = \mathbb{E}\left[\left(a(X-\mu_1) + b(Y-\mu_2)\right)^2\right]$

$= \mathbb{E}\left[a^2(X-\mu_1)^2\right] + \mathbb{E}\left[b^2(Y-\mu_2)^2\right]$

(Cross term = 0, due to independence)

$\Rightarrow \mathrm{Var}(X') = (a^2 + b^2)\sigma^2$

Similarly,

$$\mathrm{Var}(Y') = (c^2 + d^2)\sigma^2$$

- $\mathrm{Cov}(X', Y') =$

$$\mathbb{E}\left[\left(a(X-\mu_1) + b(Y-\mu_2)\right)\left(c(X-\mu_1) + d(Y-\mu_2)\right)\right]$$

$$= ac\sigma^2 + bd\sigma^2 = (ac + bd)\sigma^2$$

(Again, cross terms have zero expectation due to the independence of X and Y)

So, the covariance Matrix is

$$\sigma^2 \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix}$$

- Pearson Correlation coefficient ...

$$\rho(X', Y') = \frac{\mathrm{Cov}(X', Y')}{\sqrt{\mathrm{Var}(X')\,\mathrm{Var}(Y')}} = \frac{ac + bd}{\sqrt{(a^2 + b^2)(c^2 + d^2)}}$$